T3 group factors: EFT_1_after_F

Active theory: SM + S_1 + S_2 + $C_{LLS_1S_2}$. The fermion F is absent, so y_1, y_2 and M_F are not running EFT couplings.

$$j = \frac{d-1}{2}, \qquad C_2(d) = \frac{d^2-1}{4}, \qquad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

Notation key

symbol	interaction / definition
g_1, g_2, g_3	SM gauge interactions
Y_u, Y_d, Y_e	SM Yukawa interactions
y_1	LFS_1 + h.c.
y_2	LFS_2 + h.c.
M_F	$-\bar{F}M_F F$
T	$\mathrm{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\mathrm{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\mathrm{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2, m_2^2	$m_1^2 S_1^\dagger S_1,\ m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_i}^{(1)}$	$\frac{1}{2}\lambda_{S_i}^{(1)} (S_i^\dagger S_i)^2$
$\lambda_{HS_i}^{(1)}$	$(H^\dagger H)(S_i^\dagger S_i)$
$\lambda_{HS_i}^{(A)}$	$(H^\dagger T^A H)(S_i^\dagger T^A S_i)$
λ_5	$HHS_1S_2^\dagger$ + h.c.
$\lambda_{12}^{(1)}$	$(S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$(S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$(S_1^\dagger S_2)(S_2^\dagger S_1)$

Stage field content

run	S_1	S_2	F	integrated at this stage	active after matching
T3-C	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	F	S1, S2

Field representations

run	S_1	S_2	α
$\mathrm{T3-C}$	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	-1

Gauge and Yukawa beta functions in this EFT

$\beta_{g_1}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-C}$ $(\alpha=-1)$
g_1^3	$\frac{43}{6}$

$\beta_{g_2}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-C}\ (\alpha=-1)$
g_2^3	$-\frac{17}{6}$

$\beta_{g_3}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-C}\ (\alpha=-1)$
g_3^3	-7

$\beta_{Y_u}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-C}\ (\alpha=-1)$
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1

$\beta_{Y_d}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-C}\ (\alpha=-1)$
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1

$\beta_{Y_e}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-C}\ (\alpha=-1)$
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1

$\beta_{m_1^2}$

$\mathcal{S}$	$\mathrm{T3-C}\ (\alpha=-1)$
$\lambda_{S_1}^{(1)}\, m_1^2$	6
$g_2^2\, m_1^2$	$-\frac{9}{2}$
$g_1^2\, m_1^2$	$-\frac{3}{2}$
$\lambda_{12}^{(1)}\, m_2^2$	4

$\beta_{m_2^2}$

$\mathcal{S}$	$\mathrm{T3-C}\ (\alpha=-1)$
$\lambda_{S_2}^{(1)}\, m_2^2$	6
$g_2^2\, m_2^2$	$-\frac{9}{2}$
$g_1^2\, m_2^2$	$-\frac{3}{2}$
$\lambda_{12}^{(1)}\, m_1^2$	4

$\beta_{\lambda_{HS_1}^{(1)}}$

$\mathcal{S}$	$\mathrm{T3-C}\ (\alpha=-1)$
$\lvert\lambda_5\rvert^2$	6
$g_2^2\, \lambda_{HS_1}^{(1)}$	-9
$g_1^2\, \lambda_{HS_1}^{(1)}$	-3
$\lambda_1\, \lambda_{HS_1}^{(1)}$	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4
$\lambda_{HS_1}^{(1)}\, \lambda_{S_1}^{(1)}$	6
$\lambda_{12}^{(1)}\, \lambda_{HS_2}^{(1)}$	4
g_2^4	$\frac{9}{4}$
g_1^4	$\frac{3}{4}$
$\left(\lambda_{HS_1}^{(A)}\right)^2$	$\frac{3}{4}$
$\lambda_{HS_1}^{(1)}\, T$	2

$\beta_{\lambda_{HS_2}^{(1)}}$

$\mathcal{S}$	$\mathrm{T3-C}\ (\alpha=-1)$
$\lvert\lambda_5\rvert^2$	6
$g_2^2\, \lambda_{HS_2}^{(1)}$	-9
$g_1^2\, \lambda_{HS_2}^{(1)}$	-3
$\lambda_1\, \lambda_{HS_2}^{(1)}$	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4
$\lambda_{HS_2}^{(1)}\, \lambda_{S_2}^{(1)}$	6
$\lambda_{12}^{(1)}\, \lambda_{HS_1}^{(1)}$	4
g_2^4	$\frac{9}{4}$
g_1^4	$\frac{3}{4}$
$\left(\lambda_{HS_2}^{(A)}\right)^2$	$\frac{3}{4}$
$\lambda_{HS_2}^{(1)}\, T$	2

$\beta_{\lambda_{12}^{(1)}}$

$\mathcal{S}$	$\mathrm{T3-C}\ (\alpha=-1)$
$\lvert\lambda_5\rvert^2$	3
$g_2^2\, \lambda_{12}^{(1)}$	-9
$g_1^2\, \lambda_{12}^{(1)}$	-3
$\left(\lambda_{12}^{(1)}\right)^2$	4
$\lambda_{12}^{(1)}\, \lambda_{S_1}^{(1)}$	6

$\lambda_{12}^{(1)}\,\lambda_{S_2}^{(1)}$	6
$\lambda_{HS_1}^{(1)}\,\lambda_{HS_2}^{(1)}$	4
g_2^4	$\frac{9}{4}$
g_1^4	$\frac{3}{4}$
$\left(\lambda_{12}^{(A)}\right)^2$	$\frac{3}{4}$

β_{λ_5}

$\mathcal{S}$	$\mathrm{T3-C}\ (\alpha=-1)$
$\lambda_5\,\lambda_1$	2
$\lambda_5\,\lambda_{HS_1}^{(1)}$	4
$\lambda_5\,\lambda_{HS_2}^{(1)}$	4
$\lambda_5\,\lambda_{12}^{(1)}$	2
$g_2^2\,\lambda_5$	-9
$g_1^2\,\lambda_5$	-3
$\lambda_5\,\lambda_{HS_1}^{(A)}$	-2
$\lambda_5\,\lambda_{HS_2}^{(A)}$	2
$\lambda_5\,\lambda_{12}^{(A)}$	$-\frac{1}{2}$
$\lambda_5\,T$	2

Additional scalar-coupling beta functions

$\beta_{\lambda_{HS_1}^{(A)}}$

$\mathcal{S}$	$\mathrm{T3-C}\ (\alpha=-1)$
$g_1^2 g_2^2$	-6

$\beta_{\lambda_{HS_2}^{(A)}}$

$\mathcal{S}$	$\mathrm{T3-C}\ (\alpha=-1)$
$g_1^2 g_2^2$	6

$\beta_{\lambda_{12}^{(A)}}$

$\mathcal{S}$	$\mathrm{T3-C}\ (\alpha=-1)$
$g_1^2 g_2^2$	-6

Full Weinberg-coefficient RGE in EFT1

Writing the canonical mixed LLSS coefficient as C_{12}, the coupled one-loop Weinberg equation used in the F-first EFT is

$$16\pi^2\frac{dC_5}{d\ln\mu} = (2\lambda_1 - 3g_2^2 + 2T)C_5 \tag{1}$$

$$-\frac{3}{2}\left[(Y_e^\dagger Y_e)^T C_5 + C_5(Y_e^\dagger Y_e)\right] \tag{2}$$

$$+ R_W\,\lambda_5\,C_{12}. \tag{3}$$

Note: $\frac{7}{2}$ thing isnt there if we integrate out our fermion F so its not here

Direct $C_{12} \to C_5$ recoupling factor

For each representation the source term is

$$16\pi^2\,\beta_{C_5} \supset R_W\,\lambda_5\,C_{12}.$$

$\mathcal{S}$	$\mathrm{T3-C}\ (\alpha=-1)$
R_W	$-\frac{4}{3}$

.

$\mathrm{T3-C}\ (\alpha=-1)$

4 independent complex matched structures displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j)\,S_1^\dagger\,S_1^\dagger + \mathrm{h.c.}$	$\frac{\overline{y_1}^2}{2M_F}$	cg2
$(L_i^T C L_j)\,S_1^\dagger\,S_2 + \mathrm{h.c.}$	$-\frac{\sqrt{3}\overline{y_1 y_2}}{2M_F}$	ϵ
$(L_i^T C L_j)\,S_1^\dagger\,S_2 + \mathrm{h.c.}$	$-\frac{\overline{y_1 y_2}}{2M_F}$	$\mathcal{I}_{LLS_1S_2}$
$(L_i^T C L_j)\,S_2\,S_2 + \mathrm{h.c.}$	$\frac{\overline{y_2}^2}{2M_F}$	cg1

Next threshold

Further matching depends on the next field or fields selected by the threshold plan. Once all T3 fields are removed, their representation dependence is carried by the matched C_5 boundary condition and the subsequent SMEFT running is universal.